

ARAD, Romania

WMO: 152000

Lat: 46.1336N

Lon: 21.3536E

Elev: 118

StdP: 99.92

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.1	-14.1	1.1	-11.0	-11.4	1.4	-8.1	9.6	0.8	7.9	0.6	1.4	0	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	34.3	21.1	32.7	20.9	30.9	20.4	23.0	30.6	22.0	29.7	21.2	28.4	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.7	25.8	19.9	14.8	24.6	18.9	13.9	23.6	68.6	30.1	64.8	29.1	61.8	28.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.3	5.2	-16.5	36.5	4.4	1.9	-19.7	37.8	-22.2	38.9	-24.7	40.0	-27.9	41.4		
			-16.7	25.0	4.3	1.7	-19.8	26.2	-22.4	27.2	-24.8	28.2	-27.9	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	0.0	1.8	6.3	12.0	16.8	20.4	22.3	22.4	17.4	12.0	6.7	1.2
	DBStd	9.01	4.75	5.42	4.59	3.92	3.63	3.52	3.22	3.34	3.81	4.49	4.66	4.87
	HDD10.0	1118	311	229	129	24	1	0	0	0	32	118	273	
	HDD18.3	2834	569	462	373	190	73	19	4	6	61	199	348	531
	CDD10.0	1728	0	1	15	86	212	312	381	383	223	94	21	1
	CDD18.3	404	0	0	0	1	26	81	128	131	34	3	0	0
	CDH23.3	4578	0	0	0	37	322	899	1445	1459	374	41	1	0
	CDH26.7	1849	0	0	0	4	69	339	650	670	114	3	0	0
Wind	WSAvg	2.0	2.1	2.3	2.5	2.3	2.1	1.8	1.8	1.7	1.9	1.9	2.2	2.1
Precipitation	PrecAvg	629	35	35	37	50	62	81	70	56	55	52	47	47
	PrecMax	844	70	90	138	131	146	181	160	158	140	110	113	104
	PrecMin	292	7	1	4	3	17	21	22	3	9	1	0	3
	PrecStd	132	17	23	28	28	29	39	39	36	28	31	26	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	16.4	21.1	26.9	30.2	34.1	36.9	36.9	33.0	26.6	20.9	14.0
		MCWB	9.3	10.6	12.5	17.0	19.1	21.8	21.3	21.3	19.5	17.0	14.3	10.3
	2%	DB	9.9	13.4	18.5	24.1	28.6	32.2	34.3	34.6	29.6	24.3	17.9	11.4
		MCWB	7.6	9.0	11.2	15.2	18.6	21.5	21.4	20.8	18.3	16.0	12.7	8.8
	5%	DB	8.1	11.0	16.1	21.9	26.9	30.5	32.5	32.7	27.4	22.1	15.8	9.4
		MCWB	6.4	7.9	10.2	13.8	17.9	20.7	21.0	20.8	17.9	15.3	11.7	7.7
	10%	DB	6.2	8.9	13.9	19.9	25.0	28.5	30.3	30.6	25.2	19.5	13.7	7.6
		MCWB	5.1	6.7	9.0	12.8	17.1	19.8	20.4	20.2	17.1	14.3	10.8	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	11.1	13.8	17.7	20.8	24.4	24.2	23.9	20.8	18.2	14.8	10.9
		MCDB	12.0	15.0	19.5	24.8	27.4	30.9	32.2	31.7	29.2	24.3	19.1	13.3
	2%	WB	7.9	9.6	12.0	16.0	19.6	22.7	22.9	22.6	19.5	17.0	13.5	9.3
		MCDB	9.5	13.0	16.9	22.6	26.6	29.7	30.6	30.8	26.9	22.1	16.9	11.1
	5%	WB	6.5	8.3	10.7	14.9	18.6	21.5	21.8	21.6	18.6	15.9	12.2	7.8
		MCDB	7.9	10.6	15.2	20.6	25.2	28.2	29.5	29.6	25.2	20.6	15.2	9.3
	10%	WB	5.1	6.9	9.5	13.6	17.6	20.5	21.0	20.7	17.7	14.8	10.9	6.4
		MCDB	6.1	8.7	13.4	18.5	23.7	26.8	28.5	28.1	23.5	19.1	13.4	7.5
Mean Daily Temperature Range	5% DB	MDBR	6.3	8.0	10.4	11.8	12.3	12.3	13.1	13.3	11.8	11.2	8.6	5.9
		MCDBR	8.4	11.4	14.1	15.5	15.3	15.3	16.0	16.2	15.3	14.1	11.4	8.1
		MCWBR	6.2	7.7	8.2	8.0	7.1	6.5	5.7	5.7	6.1	7.2	7.1	6.1
	5% WB	MCDBR	7.9	10.1	12.4	13.7	13.9	13.9	14.3	14.1	13.2	12.5	10.6	7.6
		MCWBR	6.1	7.0	7.6	7.5	7.0	6.4	6.0	6.0	6.0	6.9	7.0	6.0
Clear-Sky Solar Irradiance	taub	0.339	0.365	0.402	0.458	0.440	0.437	0.450	0.456	0.430	0.390	0.361	0.339	
	taud	2.330	2.267	2.201	2.058	2.174	2.220	2.185	2.171	2.209	2.310	2.346	2.347	
	Ebn at Noon	740	792	814	799	830	833	816	793	777	754	707	695	
	Edh at Noon	78	100	124	156	143	138	141	138	122	95	77	70	
All-Sky Solar Radiation	RadAvg	1.13	1.98	3.25	4.64	5.62	6.16	6.11	5.48	3.83	2.57	1.50	0.89	
	RadStd	0.12	0.29	0.43	0.44	0.52	0.48	0.47	0.51	0.42	0.27	0.13	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	-120	N/A	+99	N/A
(47)	Regional (63 neighbors)	+0.69	N/A	N/A	N/A	+0.38	N/A	-122	-184	+116	+59

Nomenclature: See separate page